

[Dashboard](#) / [My courses](#) / [COMS3005A-AAA3-S2-2024](#) / [Quizzes](#) / [Quiz 1 for 2024](#)**Started on** Friday, 23 August 2024, 7:07 PM**State** Finished**Completed on** Friday, 23 August 2024, 7:19 PM**Time taken** 12 mins 30 secs**Grade** 18.00 out of 20.00 (90%)Question **1**

Correct

Mark 1.00 out of 1.00

A pushdown automaton is more powerful than a finite automaton because it has a stack that can be used as limited memory.

Select one:

☒ True ✓☐ False

The correct answer is 'True'.

Question **2**

Correct

Mark 1.00 out of 1.00

A Turing Machine accepts

- ☐ a. non context free languages
- ☐ b. context free languages
- ☐ c. regular languages
- ☐ d. regular languages and context free languages
- ☒ e. regular, context free and non context free languages ✓

Your answer is correct.

The correct answer is:

regular, context free and non context free languages

Question **3**

Correct

Mark 1.00 out of 1.00

A Universal Turing Machine can execute any other Turing Machine.

Select one:

☒ True ✓

☐ False

One simply needs to encode the other TM into the correct format for the UTM

The correct answer is 'True'.

Question **4**

Correct

Mark 1.00 out of 1.00

"Any problem for which there exists an algorithmic solution can be solved by a Turing machine."

or, stated differently,

"Every effectively calculable function is computable by some Turing machine."

This claim is called the ✓ .

The correct answer is: Church-Turing Thesis

Question **5**

Correct

Mark 1.00 out of 1.00

A problem that can be solved in a "reasonable time" is called a ✓ problem.

A one word answer only!

The correct answer is: tractable

Question **6**

Correct

Mark 1.00 out of 1.00

The execution of a set of instructions by an agent is called a ✓ .

A one word answer only!

The correct answer is: computation

Question **7**

Correct

Mark 1.00 out of 1.00

A function is *computable* (also called *effectively calculable* or simply *calculable*) if it can be calculated by a finite mechanical procedure.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **8**

Correct

Mark 1.00 out of 1.00

Turing Machines are an important theoretical concept because

- ☒ a. they are a model of a general purpose computer ✓
- ☐ b. they can be used to solve any problem
- ☐ c. they are powerful machines
- ☐ d. they can execute programs as well as accept languages
- ☐ e. they are more powerful than push down automata

Your answer is correct.

The correct answer is:
they are a model of a general purpose computer

Question 9

Correct

Mark 1.00 out of 1.00

Every quintuple in a Turing Machine program must be defined uniquely by its first two symbols.

Select one:

☒ True ✓

☐ False

A basic Turing Machine is deterministic therefore we cannot have choices as to what do to when a particular input symbol is read while in a particular state.

The correct answer is 'True'.

Question 10

Correct

Mark 1.00 out of 1.00

The Halting problem is ✓ .

A one word answer only!

The correct answer is: undecideable

Question 11

Incorrect

Mark 0.00 out of 1.00

The Halting Problem is a important result in computability theory because it can be used to show that

Answer: ✗

The correct answer is: other problems are undecideable

Question **12**

Correct

Mark 1.00 out of 1.00

A *Basic Operation* in complexity theory is:

- ☐ a. All of the operations (e.g. +, -, /, etc.) performed by an algorithm in solving the problem
- ☐ b. Assignment statement
- ☐ c. Checking logical conditions (e.g. <, >, =, etc.)
- ☐ d. Any operation related to the problem the algorithm is trying to solve
- ☒ e. An operation that is core to the function of the algorithm ✓

Your answer is correct.

The correct answer is:

An operation that is core to the function of the algorithm

Question **13**

Correct

Mark 1.00 out of 1.00

In determining the *Space Complexity* of some algorithm we consider the memory needed to store all the data needed to solve the problem.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question 14

Correct

Mark 1.00 out of 1.00

If $f(n) = 34n^2 + 20n + 14$

- ☐ a. $f(n) \in O(n^2)$
- ☐ b. $f(n) \in o(n^2)$
- ☐ c. $f(n) \in \omega(n^2)$
- ☒ d. $f(n) \in \Theta(n^2)$ ✓
- ☐ e. All of the other answers

Your answer is correct.

The correct answers are:

$f(n) \in O(n^2)$

$f(n) \in o(n^2)$

$f(n) \in \omega(n^2)$

$f(n) \in \Theta(n^2)$

All of the other answers

Question 15

Correct

Mark 1.00 out of 1.00

If $f(n) = n^2$ then $f(n) \in o(n^2)$

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question 16

Correct

Mark 1.00 out of 1.00

If $f(n) = 13n^2 + 12n + 34$ then $f(n) \in O(n)$

Select one:

- ☐ True
- ☒ False ✓

Can't find any value of c and n_0 to satisfy the inequality

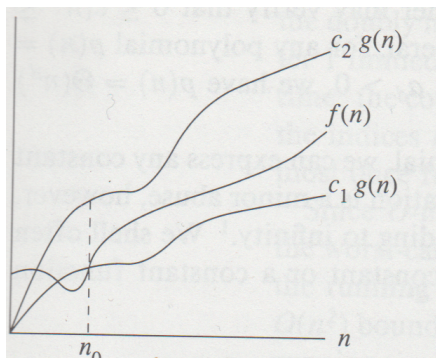
The correct answer is 'False'.

Question 17

Incorrect

Mark 0.00 out of 1.00

What does the figure below **best represent** in terms of asymptotic analysis of algorithms?



- ☐ a. $f(n) \in O(g(n))$
- ☐ b. $f(n) \in \Theta(g(n))$
- ☐ c. $f(n) \in \Omega(g(n))$
- ☐ d. $f(n) \in \omega(g(n))$
- ☒ e. All the other answers ✗

Your answer is incorrect.

The correct answer is:

$$f(n) \in \Theta(g(n))$$

Question **18**

Correct

Mark 1.00 out of 1.00

A decision problem is one that has a "yes" or "no" answer

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **19**

Correct

Mark 1.00 out of 1.00

The Minimum Vertex Cover Problem is an optimisation problem whose decision problem falls in the class of NP-complete ✓

The correct answer is: NP-Complete

Question **20**

Correct

Mark 1.00 out of 1.00

The *Travelling Salesperson problem* has a *polynomial* time complexity.

Select one:

- ☐ True
- ☒ False ✓

It is generally considered to have only an exponential time complexity. If we could solve this problem in polynomial time then we could solve many others (all the NP-Complete problems) in polynomial time too.

The correct answer is 'False'.

◀ [Red Black Tree slides](#)

Jump to...

[Quiz 2 for 2024 -- AI search ▶](#)